The Boson/Fermion Distinction: Spin-Statistics as Structural Invariant

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Abstract

The textbook dichotomy between bosons and fermions is commonly presented as a primitive classification of nature, with the spin-statistics theorem as its iron law. This paper argues that the dichotomy is a derived, dimension-dependent shadow of a single structural relation: the exchange phase of identical particles equals their topological spin, $R = e^{2\pi i s}$. [ESTABLISHED] The relation holds across relativistic quantum field theory, topological field theory, and condensed-matter anyon systems; dimension enters only by quantizing the allowed values of s. [RETRODICTION — not evidence] Stating this as a single invariant is a unification of established results, not a new prediction. After stating the invariant and its dimensional quantization, the paper addresses a foundational question: whether a calculus whose primitive is the distinction — rather than the particle or the field — can derive exchange statistics from the act of distinction itself. A recent monograph in this tradition (Quni-Gudzinas, 2026a) exhibits the gap [textual finding]: the model of its exponential modality silently adopts the symmetric algebra, which corresponds to bosonic statistics, without deriving that choice from the primitive. The paper formalizes the required construction — two modal exponentials (symmetric and exterior), the braiding of two marks in a compact closed category, and the ribbon condition linking twist to exchange — and states the falsifiability conditions under which the derivation program succeeds or fails. [NOT YET EVIDENCE] The derivation is pre-registered here; it is not yet executed.

1. Introduction

Two identical particles can be exchanged. In quantum mechanics, the state of the pair carries a phase under exchange: $\psi(x_2, x_1) = \eta\,\psi(x_1, x_2)$. In three or more spatial dimensions, exchanging twice is topologically trivial, so $\eta^2 = 1$ and $\eta \in \{+1, -1\}$: the two possibilities are bosons (symmetric, $\eta = +1$) and fermions (antisymmetric, $\eta = -1$). The spin-statistics theorem states which sign is realized: $\eta = (-1)^{2s}$, where s is the spin (Pauli, 1940; Duck and Sudarshan, 1998). [ESTABLISHED]

This paper makes two claims. First, the structural invariant is not the dichotomy itself but the relation $R = e^{2\pi is}$ between exchange phase and topological spin; the binary is a shadow of that relation in three spatial dimensions. Second, a distinction-based foundation of physics — a calculus whose only primitive is the act of drawing a boundary — must, to claim the spin-statistics connection, derive exchange statistics from the primitive; the current state of that program contains a silent assumption that this paper identifies and replaces with a concrete derivation target.

2. The invariant: exchange phase equals topological spin

The exchange of two identical particles is a loop in their configuration space. The phase acquired is a representation of the fundamental group of that space (Leinaas and Myrheim, 1977). [ESTABLISHED] The rotation of a single particle by 2π is the twist. In a ribbon braided tensor category — the mathematical home of particle-like excitations in topological order — the two are linked by the ribbon identity:

$$\theta_X = \frac{\mathrm{Tr}_q(c_{X,X})}{d_X},$$

where θ_X is the twist (topological spin), $c_{X,X}$ the braiding, d_X the quantum dimension, and Tr_q the quantum trace (Joyal and Street, 1993; Kitaev, 2006; Bakalov and Kirillov, 2001). [ESTABLISHED] For an abelian object, $d_X = 1$ and the identity reduces to

$$R = e^{2\pi is},$$

the universal spin-statistics relation. This relation has been proven directly from wavefunctions for fractional quantum Hall quasiparticles (Trung et al., 2022; Nardin et al., 2022) and is realized experimentally as measurable fractional spin (Comparin et al., 2021). [ESTABLISHED]

2.1 The 3+1D shadow

In three or more spatial dimensions the braiding is symmetric (involutive): $c_{Y,X} \circ c_{X,Y} = \mathrm{id}$. Then $\theta_X^2 = 1$, so $e^{2\pi is} = \pm 1$, forcing $s \in \{0, \frac{1}{2}\} \bmod 1$. Integer spin gives the trivial representation (bosons); half-integer spin gives the sign representation (fermions) (Pauli, 1940; Streater and Wightman, 1964). [ESTABLISHED]

2.2 The 2+1D generalization

In two spatial dimensions the braiding is not involutive: the exchange group is the braid group, and $\theta_X = e^{2\pi is}$ can be any phase. Particles with fractional statistics — anyons — realize the continuous range of $s \in \mathbb{R}/\mathbb{Z}$ (Leinaas and Myrheim, 1977; Wilczek, 1982; Mund, 2008). [ESTABLISHED]

2.3 What is invariant

Across both regimes, the invariant content is unchanged: superselection sectors, fusion rules, and braiding data, with the relation $R = e^{2\pi is}$ (Wang and Wen, 2014; Johnson-Freyd, 2015). [ESTABLISHED] Dimension enters only through which braided structures are realizable. The binary dichotomy is therefore not the invariant; the relation is.

3. Dimension quantization: bosons and fermions as a 3+1D shadow

Spatial dimension	Motion group	Braided structure	Allowed s	Statistics
2	Braid group B_n	Ribbon, non-symmetric	$s \in \mathbb{R}/\mathbb{Z}$	Anyons, $R = e^{2\pi is}$
≥ 3	Permutation group S_n	Symmetric, involutive	$s \in \{0, \frac{1}{2}\} \bmod 1$	Bosons ($\eta = +1$), fermions ($\eta = -1$)

[ESTABLISHED] The table is the dimensional quantization of the spin parameter: in $d \geq 3$ the involutive braiding forces $2s \in \mathbb{Z}$; in $d = 2$ the quantization collapses to continuity. In $3{+}1$ dimensions a further input — microcausality and positive energy in a local relativistic theory — identifies which sign corresponds to which spin (Pauli, 1940; Duck and Sudarshan, 1998; Verch, 2001). [ESTABLISHED] That input is external to any purely algebraic derivation; the boundary is stated explicitly in Section 5.

4. The calculus of distinctions and its silent assumption

A distinction-based calculus begins with a mark: a boundary separating an inside from an outside. Its two primitive laws are Calling (idempotence: a mark repeated is the mark) and Crossing (involution: a boundary crossed twice returns to the unmarked state) (Spencer-Brown, 1969). A recent monograph develops this calculus toward physics, including a treatment of the half-turn phase $e^{i\pi} = -1$ and a claim that parity is the ancestor of physical spin-statistics (Quni-Gudzinas, 2026a, Section 2.3). [textual finding]

The monograph's Appendix A models its exponential modality with the symmetric algebra:

$$!A = \bigoplus_{n=0}^{\infty} S^n(A),$$

where $S^n(A)$ is the n-th symmetric power. The symmetric algebra is exactly the bosonic Fock construction: many-body states totally symmetric under exchange. [ESTABLISHED — the symmetric algebra realizes Bose statistics.] The monograph therefore *silently chooses bosonic statistics* as the semantic realization of its exponential, without deriving the choice from the act of distinction. [textual finding — verifiable against Quni-Gudzinas (2026a), Appendix A.]

The gap is structural, not cosmetic. The spin-statistics theorem is about the symmetry of the joint state of two identical marks under exchange; the calculus provides a phase for a single mark under rotation, but never constructs the exchange of two marks. The leap from "the mark has a half-turn phase" to "two marks anticommute" is asserted, not derived. (This assessment was reached independently in the deep-inquiry analysis of 2026-08-14; it is stated here as a checkable textual claim about the monograph.)

5. A derivation program

The program has three tasks, pre-registered with falsifiability conditions (Section 6).

T1 — Two modal exponentials. In a $\mathbb{Z}/2$-graded symmetric monoidal category with the graded braiding $\sigma_{A,B}(a \otimes b) = (-1)^{|a||b|} b \otimes a$, the exchange operator $P = \sigma_{A,A}$ on $A \otimes A$ splits into two idempotent projectors,

$$P_{\text{sym}} = \tfrac{1}{2}(1 + P), \qquad P_{\text{antisym}} = \tfrac{1}{2}(1 - P),$$

whose eigenspaces are the symmetric and antisymmetric subspaces. [ESTABLISHED — elementary super-algebra.] For a mark of odd parity, the graded symmetric algebra coincides with the exterior algebra, so the symmetric and exterior exponentials,

$$!_S(A) = \bigoplus_n S^n(A), \qquad !_\Lambda(A) = \bigoplus_n \Lambda^n(A),$$

are the two grading components of one construction. The two statistics are the two eigenvalues of exchange. [DERIVATION SKETCH — P4 notebook T1.]

T2 — The braiding of two marks. In a compact closed category with a self-dual mark M, the exchange map $\sigma_{M,M}$ is a scalar $\eta \cdot \mathbf{id}$, and the ribbon identity forces $\eta = \theta_M$. In a symmetric category $\theta_M^2 = 1$, so $\eta = \pm 1$: the two eigenvalues of exchange are the boson and fermion signs. The sign $\eta = -1$ is the same -1 as the treatise's half-turn phase $e^{i\pi} = -1$. [DERIVATION SKETCH — P4 notebook T2.] The identification of $\eta = +1$ with Calling and $\eta = -1$ with Crossing is the mark-calculus reading of the two one-dimensional representations of the symmetric group.

T3 — Dimension quantization. The table in Section 3, with dimension entering only through the allowed braided structures. [DERIVATION SKETCH — P4 notebook T3.]

The boundary of the program. The program can show that statistics is forced by distinction, compact closure, and an involutive braiding (the two eigenvalues of exchange). It cannot, from the mark alone, show *which* eigenvalue corresponds to *which* spin: the spin–statistics *connection* requires the additional postulate that the twist equals the 2π rotation of a Lorentz representation, with microcausality and positive energy (Pauli, 1940; Duck and Sudarshan, 1998). The paper states this boundary explicitly; it does not claim a full derivation of the spin-statistics theorem from distinction. [CONTESTED — the sufficiency of the minimal postulates is open.]

6. Falsifiability conditions

F1 (empirical). If a stable, local, relativistic excitation in $3+1$ dimensions is observed with exchange phase $\eta \neq e^{2\pi i s}$ — for example, a spin-$\frac{1}{2}$ particle obeying Bose-Einstein statistics, or a spin-0 particle obeying Fermi-Dirac statistics — the claim that $R = e^{2\pi i s}$ is the universal invariant is disconfirmed. No such particle is known in the Standard Model. [ESTABLISHED — the absence of violations is a strong constraint, not a proof.]

F2 (formal). If the mark calculus cannot reproduce the two one-dimensional representations of S_n (trivial and sign) from the primitive distinction, compact closure, and an involutive braiding alone — without importing microcausality, Lorentz structure, or any other physical postulate — the derivation program is disconfirmed, and the monograph's spin-statistics claim stands as an asserted correspondence.

Surprise accounting (KIF-60 discipline). The existence of anyons in $2+1$ dimensions is established and does not count as predictive evidence for this paper: anyonic statistics is expected under the null hypothesis of braid-group representations. Only F1's precision constraint and F2's derivability constraint carry evidential weight. The invariant formulation itself is [RETRODICTION — not evidence]: it restates established results in a unified language. The paper claims credit only for the derivation program (F2) and for the identification of the monograph's silent assumption (a textual finding).

7. Relation to existing programs

The derivation program sits between three established lines of work. First, the algebraic quantum field theory tradition proves the spin-statistics connection from locality and positivity (Streater and Wightman, 1964; Verch, 2001), and extends it to anyons in $2+1$ dimensions (Mund, 2008; Kuckert, 2002; Kuckert and Mund, 2004). Second, the topological and categorical tradition states the connection as a theorem about braided tensor categories (Joyal and Street, 1993; Bakalov and Kirillov, 2001; Johnson-Freyd, 2015; Oeckl, 2000), and condensed-matter physics realizes it for fractional quantum Hall quasiparticles (Comparin et al., 2021; Nardin et al., 2022; Trung et al., 2022). Third, the distinction-based tradition derives physical structure from the mark (Spencer-Brown, 1969; Quni-Gudzinas, 2026a, 2026b, 2026c). The present paper is the first, to the author's knowledge, to state the derivation target explicitly for the third tradition: the exchange of two marks, constructed in a compact closed category, whose eigenvalues are the two statistics. Adjacent internal work on p-adic anyon braiding (Quni-Gudzinas, 2026d) and on the topological distinction between Dirac and Majorana fermions (Quni-Gudzinas, 2026e) provides a compatible categorical language.

8. Conclusions

The boson/fermion dichotomy is not the primitive content of the spin-statistics theorem; the relation $R = e^{2\pi i s}$ between exchange phase and topological spin is. [ESTABLISHED] The dichotomy is its shadow in three spatial dimensions, where the involutive braiding quantizes s to integers and half-integers. [ESTABLISHED] A distinction-based calculus that aims to ground quantum statistics must construct the exchange of two marks and derive the two eigenvalues of exchange from the primitive; the current monograph in that tradition silently assumes the symmetric (bosonic) algebra instead. [textual finding] This paper pre-registers the derivation program (T1-T3) and its falsifiability conditions (F1, F2), and states the boundary of the program: the spin-statistics *connection* requires Lorentz and locality input that the mark alone cannot supply. [NOT YET EVIDENCE]

Declarations

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